

CS 271

Systems in C

Calvin Deutschbein

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Office Hours: TTh 1110-1310

Office: Ford 307, Discord

cd-public.github.io/

Lecture: TTh 1310-1440

Classroom: Walton Hall 235

Course Description

AI-101

A semester-long study of topics in Computer Science that specifically includes a collaborative programming project that applies or extends the content of CS 152 or more advanced classes.

Systems research joins together the theory and the practice of computing, and cryptography is a perfect microcosm of this dynamic. As a guided study of low-level implementation of computer systems, this course will study the RSA and SHA cryptographic at once in theory and practice, and then build these cryptographic elements into emergent systems using data structures and algorithms such as Merkle trees and blockchains.

About Me

Calvin Deutschbein is an assistant professor of computer science.

Required Materials

Required materials for a given class will be available on the course webpage. All course materials will be made available at no cost to the student.

Accessibility

I will make every effort to ensure all coursework and materials are accessible to all students, including working with on-campus specialists. However, there is always room for improvement. I always appreciate hearing from students about how I can make the course more accessible, so please reach out if there is something I can be doing better!

Course Objectives

Upon completion of the course, students should be able to:

- Differentiate confidentiality, integrity, and availability.

- Apply modular arithmetic and other number theoretic techniques to computational systems.
- Articulate the impossibility of distributed consensus.
- Apply recursion and pointers to programming problems.
- Understand a low-level systems programming language and its uses.

This course is decomposed structurally into two modules: a module on cryptography, which explores mathematics, principles of cybersecurity, and beginning usage of a low-level systems programming language, and a module on cryptocurrency, which explores advanced algorithmic problems of recursion and distributed consensus and the applications of systems programming.

Course Structure

The course will be composed of lecture, labs, and homework.

Class Structure

Classes are scheduled twice weekly. The lecture schedule is on the course webpage.

Feedback and Grading

Grading Scale

This course will be ungraded. Systems computing is difficult and may be stressful, and I recall students struggling in courses like this because of grading policies rather than because of course content. Therefore, all work will support the eventual final, and the eventual final alone can guarantee a high grade.

- **Students will begin the course with a grade of an “A”**
 - Students will be expected to attend class.
 - Students will be expected to participate in class.
 - Students will be expected to treat fellow students with respect.
 - Students will be expected to complete response assignments.
- **Students will be contacted privately by the course instructor in the unusual event they are not meeting expectations.**
 - Students will not have an expectation of perfection.
 - Students will not lose points or a grade without discussion.
 - Students will have a chance to explain their engagement with the course.
- **Students collectively within the class as a whole will receive collective feedback on ethics, communication, and teamwork.**
 - Students will receive limited individual feedback in unique cases.
 - Students will be expected to provide respectful individual feedback to one another.
 - Students will be able to request feedback from the instructor at any time.
 - Students will receive narrative rather than quantitative feedback.

These policies apply to all coursework up to the final project, which must be completed to earn a high score (B- or higher) in the course.

A completed final is submitted on time and contains all required elements at the required length. The final is described on the course webpage “Final” page.

Course Policies

Collaboration Policy

All collaboration is by mutual enthusiastic consent. Any collaboration is permitted on homeworks and labs, none on the midterm, and partners on the final. This will be conducted via honor system unless I receive an (anonymous) student complaint.

College Policies

The following material is adapted from “Information for Syllabus” recommended language on syllabus preparation provided to instructors in the College of Arts & Sciences.

Academic Integrity

Students of Willamette University are members of a community that values excellence and integrity in every aspect of life. As such, we expect all community members to live up to the highest standards of personal, ethical, and moral conduct. Students are expected not to engage in any type of academic or intellectually dishonest practice and encouraged to display honesty, trust, fairness, respect, and responsibility in all they do. Plagiarism and cheating are especially offensive to the integrity of courses in which they occur and against the College community as a whole. These acts involve intellectual dishonesty, deception, and fraud, which inhibit the honest exchange of ideas. Plagiarism and cheating may be grounds for failure in the course and/or dismissal from the College. <http://willamette.edu/cla/catalog/policies/plagiarism-cheating.php>

Commitment to Positive Sexual Ethics

Willamette is a community committed to fostering safe, productive learning environments, and we value ethical sexual behaviors and standards. Title IX and our school policy prohibit discrimination on the basis of sex, which regards sexual misconduct — including discrimination, harassment, domestic and dating violence, sexual assault, and stalking. We understand that sexual violence can undermine students’ academic success, and we encourage affected students to talk to someone about their experiences and get the support they need.

Please be aware that as a mandatory reporter I am required to report any instances you disclose to Willamette’s Title IX Coordinator.

If you would rather share information with a confidential employee who does not have this responsibility, please contact our confidential advocate at confidential-advocate@willamette.edu. Confidential support also can be found with SARAs and at the GRAC (503-851-4245); and at WUTalk - a 24-hour telephone crisis counseling support line (503-375-5353). If you are in immediate danger, you may reach campus safety at 503-370-6911.

DACA/Undocumented Student Advocate

Willamette is committed to supporting our DACA/Undocumented students in a variety of ways. This year, Tori Ruiz is the contact person for all DACA/undocumented students can provide those students with a number of external and internal resources that are available. Her contact information is email: truiz@willamette.edu, Office: 3rd Floor UC, Phone: 503-370-6447.

Diversity and Disability Statement

Willamette University values diversity and inclusion; we are committed to a climate of mutual respect and full participation. My goal is to create a learning environment that is usable, equitable, inclusive and welcoming. If there are aspects of the instruction or design of this course that result in barriers to your inclusion or accurate assessment or achievement, please notify me as soon as possible. Students with disabilities are also encouraged to contact the Accessible Education Services office in Smullin 155 at 503-370-6737 or Accessible-info@willamette.edu to discuss a range of options to removing barriers in the course, including accommodations.

Religious Practice

Willamette University recognizes the value of religious practice and strives to accommodate students' commitment to their religious traditions whenever possible. Please let me know within the first two weeks of the semester if a conflict between holy days or other religious practice and full participation in the course is anticipated. I will do my best to work with you to determine a reasonable accommodation.

As an instructor, I will exercise my discretion to offer accommodations for conflicts after the first two weeks of the semester. You may always reach out to me, including retroactively, though the quality of the accommodation I am able to offer may improve given advanced warning!

SOAR Center Offerings: Food, Clothing, and School Materials

The Students Organizing for Access to Resources (SOAR) Center strives to create equitable access to food, professional clothing, commencement regalia, and scholarly resources for WU and Willamette Academy students. The SOAR Center is located on the Putnam University Center's third floor (in the former Women's Resource Center and across from the Harrison Conference Room). The space houses the Bearcat Pantry, Clothing Share, and First-Generation Book Drive and is maintained by committed students and staff and faculty advisers.

Trans Inclusion and Gender Justice

I am always appreciative of the opportunity to address you by your affirming name or pronoun. Please advise me of the most affirming way to address you at any time so that I may do so.

If I ever misgender you in any way, I would greatly appreciate that you let me know, in whatever manner makes you comfortable, so that I can correct that error and endeavour to repair any harm.

Mental Health

As a student, you may experience a range of challenges that can interfere with learning, such as strained relationships, increased anxiety, substance use, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may diminish your academic performance and/or reduce your ability to participate in daily activities. Willamette services are available and treatment does work. If you think you need help, please contact Bishop Health as soon as possible at <http://willamette.edu/offices/counseling/>. Crisis counseling is available 24/7 at WUTalk: 503-375-5353 and Campus Safety is available at 503-370-6911. Emergency resources are also available from the Psychiatric Crisis Center at 503-585-4949 and the National Suicide Prevention Lifeline at 1-800-273-8255.